

Late Breaking Results: A real-time diffusion-based filter for human pose estimation on edge devices

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ABSTRACT

Human Pose Estimation (HPE) is increasingly being adopted in a wide range of applications, from healthcare to Industry 5.0. To address the intrinsic inaccuracy of such CNN-based software, the current trend involves applying filtering models to refine and improve the inference results. However, state-of-the-art filtering models are computationally intensive, limiting their use in resource-constrained devices. To overcome this limitation, we propose a real-time filtering technique based on diffusion models designed specifically for edge devices. Through a micro-benchmarking phase, we analyze how the model responds to various levels of noise and select the optimal setup for specific application scenarios. Using a widely available edge device, we evaluated the model's performance on both synthetic and real noise generated by a state-of-the-art HPE system. Preliminary results demonstrate a significant improvement in real-time filtering performance with minimal computational overhead.

KEYWORDS

human pose estimation, edge computing, real-time, diffusion model

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1 INTRODUCTION AND MOTIVATION

Human Pose Estimation (HPE) aims to predict the poses of human body keypoints in videos without the need for markers. It is increasingly being adopted in different fields, from telemedicine to robotics [1]. Most HPE frameworks rely on neural networks (NN). Despite the significant advancements in NN architectures, existing frameworks often exhibit limitations in precision due to challenging human poses, occlusions, and intrinsic model inaccuracy. To address these limitations, many filtering methods have been proposed in the last years [8]. One of the most promising filtering classes is based on Denoising Diffusion Probabilistic Models (DDPM) [3]. These NN-based models have a denoising process that involves a sequence of refinement steps to gradually improve the *probabilistic distribution* of the data towards high-quality results. Standard DDPM demands significant computational resources and introduces substantial latency [2]. This limits their use in resource-constrained edge devices (e.g., Nvidia Jetson). To address this limitation, some solutions apply the Denoising Diffusion Implicit Models (DDIM)

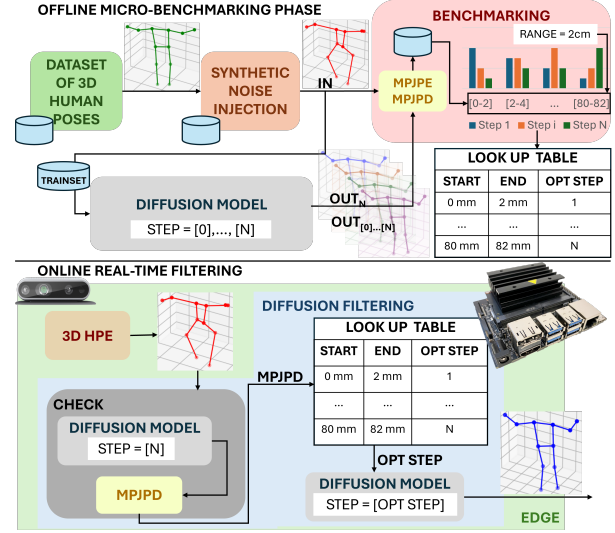


Figure 1: Methodology overview.

[7], which allows for achieving high accuracy with low computational demand by selecting a specific subset of refinement steps within the sequence. Nevertheless, the choice of the steps in the sequence and the number of steps strongly affects the refinement efficiency, with a wrong choice leading to the degradation of the HPE result. In this work, we propose a methodology to identify, in real-time, the optimal refinement step in the sequence. Through a micro-benchmarking phase conducted at the system setup time, we associate each degree of inaccuracy of the HPE data with the appropriate refinement step. At each time frame, we estimate the degree of inaccuracy and apply the corresponding step. We conducted the experimental evaluation with a widespread 3D HPE platform (TRTPose [6] with FCN[5]) and a standard dataset (Human3.6M[4]) using both synthetic and real noise.

2 METHODOLOGY

Fig. 1 presents an overview of the proposed methodology, which consists of two main phases: the *offline micro-benchmarking* and the *online real-time filtering*. The first part aims to statically analyze the behaviour of our diffusion model DM , trained using N diffusion steps, to measure how it responds using various levels of synthetic noise. To do this, we injected *additive white gaussian noise* into the dataset. This work considers this type of noise as the most standard and representative. The methodology can be applied by adopting other types of noise. In this phase, we feed a noisy pose kp_n as input to our DM , for which the actual *mean per joint position error* $mpjpe^{kp_n}$ is known. We individually apply all the steps of the reverse diffusion process, i.e., $[[N], [N-1], \dots, [0]]$, and we cache all

the filtered poses $KP_f = \{kp_f^0, \dots, kp_f^i, \dots, kp_f^N\}$ where i is the diffusion step. Then we focus on kp_f^N and compute the mean per joint position distance $mpjpd$ to kp_n . As confirmed by our experimental results, we identify a correlation between the $mpjpd$ and $mpjpe^{kp_n}$ (Fig. 2). The idea is that, as more noise is injected, the input of the noisy dataset kp_n diverges further from the output filtered kp_f^N by the DM . At this point, we determine the optimal diffusion step for this $mpjpd$ by identifying the value of i that corresponds to the minimum value in $\{mpjpe^{kp_f^i} \mid \forall i = N, N-1, \dots, 0\}$, since each diffusion step applied individually remove a different amount of noise. We apply this process in the whole dataset, and we extrapolate the best diffusion step for each range r (i.e. 20mm) of $mpjpd$, obtaining a look-up table (LUT) composed as $[start, end, opt_step]$ where $start$ and end are the start and the end of the considered range, and opt_step is the mode step extrapolated in the micro-benchmarking phase.

In the online real-time scenario, we use a 3D HPE to obtain the keypoints kp , and then, following the application of the N -th step of the diffusion model kp_f^N , we extrapolate the $mpjpd$. Using this divergence as a guiding metric, we refer to the previously acquired LUT to find the best step. Subsequently, we obtain the denoised human pose based on this decision.

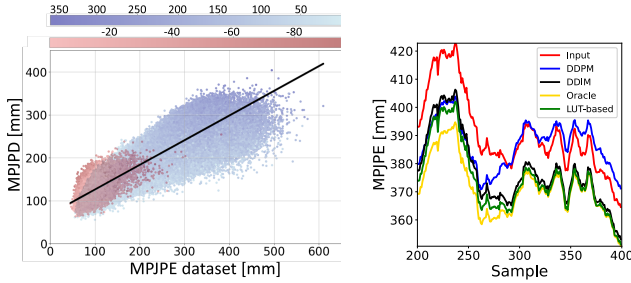


Figure 2: The left plot shows the correlation between $mpjpe$ and $mpjpd$, with their linear relation (black line). Note that there is an overlapping region between improvement (blue dots) and worsening (red dots) in accuracy. The right plot represents the test over time of a real 3D HPE.

3 EXPERIMENTAL RESULTS

We started from the implementation proposed in [2], which is a filtering DM configured with 51 diffusion steps trained on the train subjects of the Human3.6M [4] dataset. For the micro-benchmarking phase, we considered the ground truth of the *train subjects* of the dataset on which we injected synthetic noise by varying the signal-to-noise ratio, with an average MPJPE of $157 \pm 95mm$. Table 1 presents the accuracy and the performance of the following filters to refine the noisy ground truth of the *test subjects*:

- **DDPM** (51 steps): Standard *DDPM* [2].
- **DDIM** (2 steps): Diffusion model (*DM*) that applies the same fixed steps (0 and 6 of the sequence) as proposed by [2].
- **Oracle** (1 step): a *DM* filter that executes all possible steps individually and, by comparing the output to the ground truth, chooses the pose which minimizes the $mpjpe$. This approach is unfeasible in real-world domains where the ground truth is not available in real-time.
- **LUT-based** (Ours, 1 step): Using a LUT, the method determines the optimal step considering the $mpjpd$.

We found that all methods decrease the average MPJPE. The proposed filter reduces the error by 41.29%, which is even better

Filter	MPJPE (mm)	Improvement (%)	FPS
Input (Dataset)	157.23 $\pm$ 92.82	-	-
DDPM (51 steps)	95.80 $\pm$ 47.36	39.07%	0.61
DDIM (2 fixed steps)	125.31 $\pm$ 78.91	20.30%	14.02
Oracle (1 step)	86.65 $\pm$ 46.44	44.89%	-
LUT-based (Our approach, 1 step)	92.31 $\pm$ 46.58	41.29%	14.69

Table 1: Evaluation on synthetic noise

than the DDPM. The oracle represents the theoretical best result that our approach could achieve. From a performance point of view, the proposed filter outperforms the DDPM running up to 14.69 frames per second (FPS), with a speedup of 24.1 $\times$, on a Nvidia Jetson Xavier NX.

To evaluate the applicability of the proposed filter in a real-case scenario, we extrapolated 3D human poses from the dataset using a standard 2D HPE [6] with a 3D lifting neural network [5]. The plot on the right-most side of Fig. 2 shows the accuracy of the input data (HPE skeletons) and of the filtered output expressed as MPJPE over time (average of all keypoints). The results confirm that the proposed filter outperforms all the other approaches, especially when the input data is more noisy. With less noisy input data, the DDPM worsens the results, while the proposed filter behaves very similarly to the oracle. As future work we plan to investigate the efficiency of the proposed approach with industrial datasets and to improve the step selection module using machine learning techniques.

4 CONCLUSIONS

In this paper, we investigated the adoption of one-step DDIM to refine HPE data in real-time. We analysed how the model responds to different levels of noise through micro-benchmarking. The result allows us to select a specific step of the DDIM refinement sequence depending on the noise level at each time frame. We achieved an improvement in accuracy (41.29%) and a significant speedup (24.1 $\times$) on a standard dataset using synthetic noise. We also found that the proposed filter outperforms the state-of-the-art in a real case scenario.

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